Lab 5

Digital Filters

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Tasks

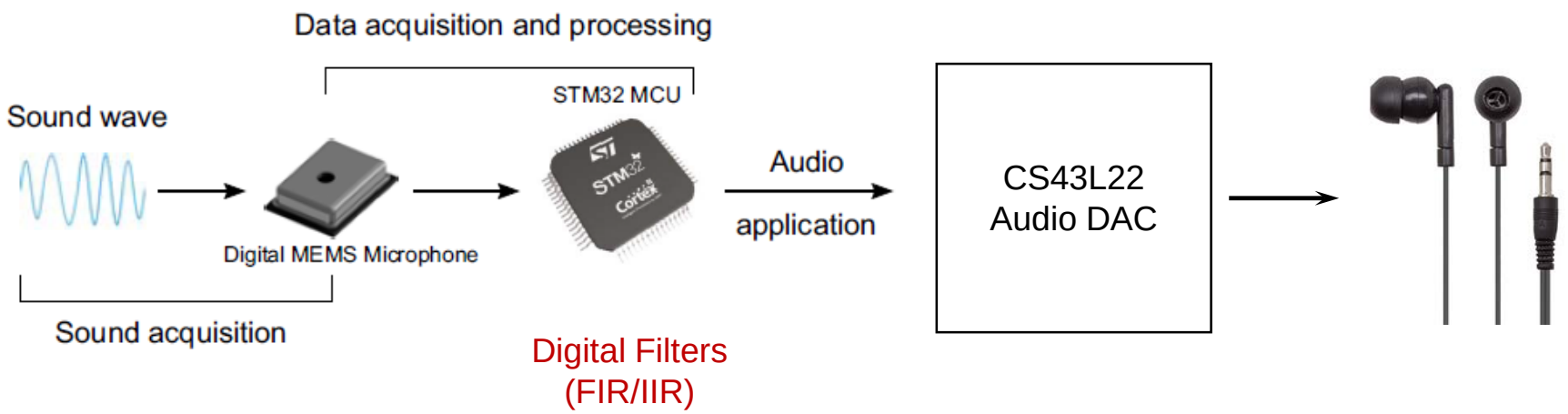
Project 1: test filter functions (C floating point, C fixed-point, hybrid inline assembly fixed-point implementations)

- IIR filter (echo)
- FIR filter
- Fixed-point implementation considerations:
  - Scaling
  - Arithmetic truncation error

Project 2: real time audio processing

- Hardware setup
- Try out filter implementations with different parameters and observe effect.
  - IIR filter (echo): vary delay and attenuation
  - FIR filter: high pass and low pass

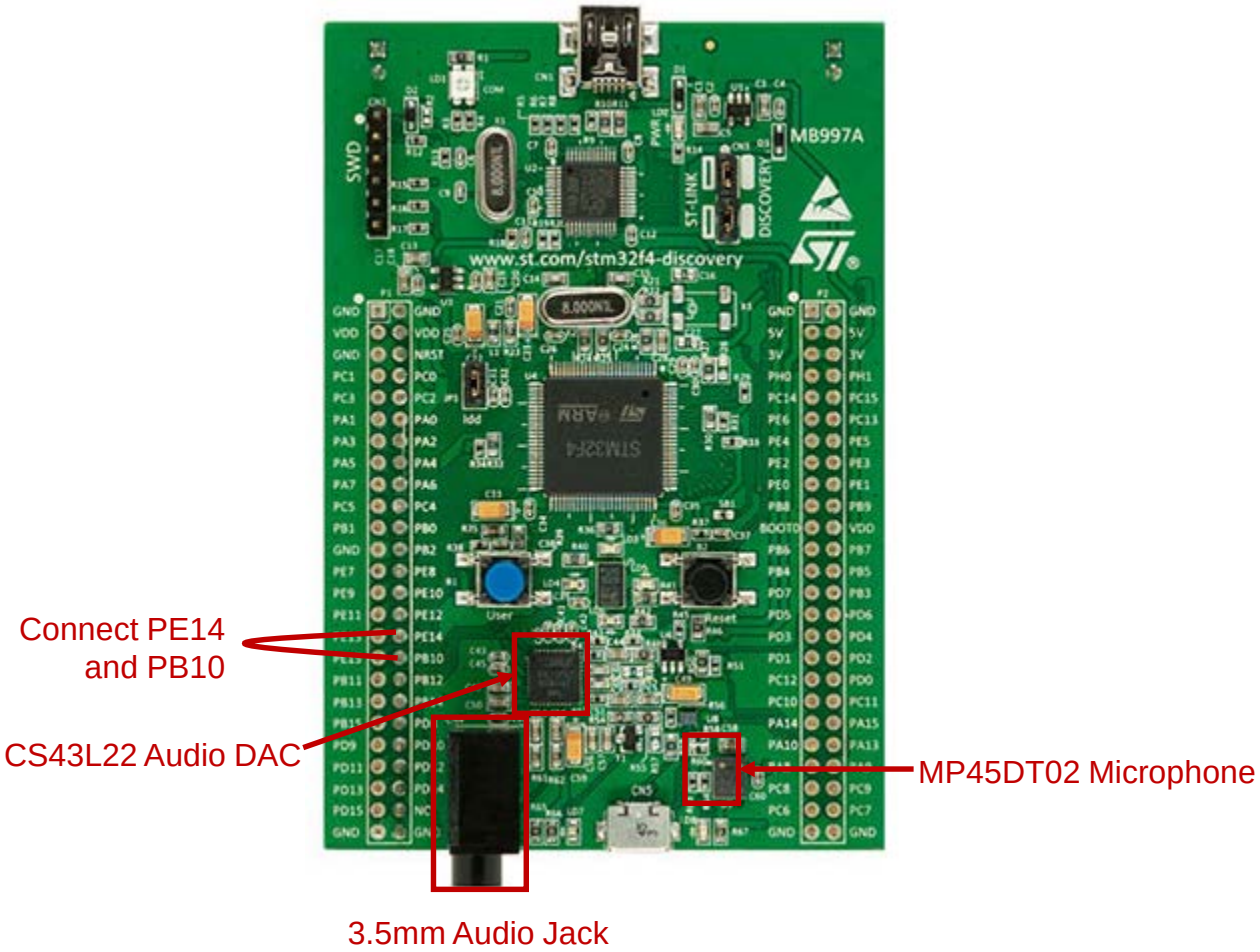
Audio Signal Processing



AN5027 Application note: Interfacing PDM digital microphones using STM32 MCUs and MPUs

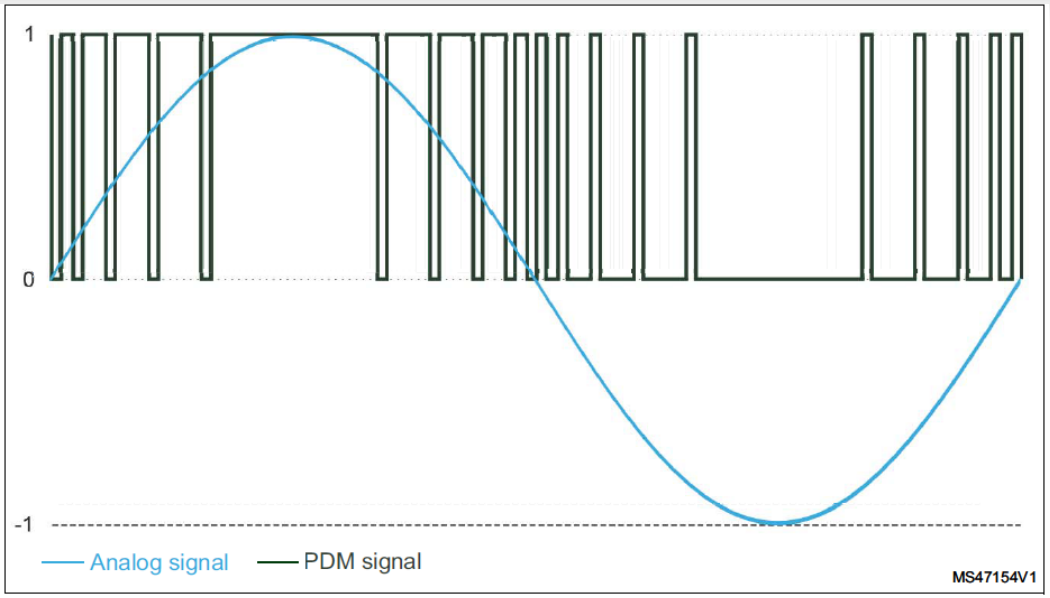
Hardware Setup

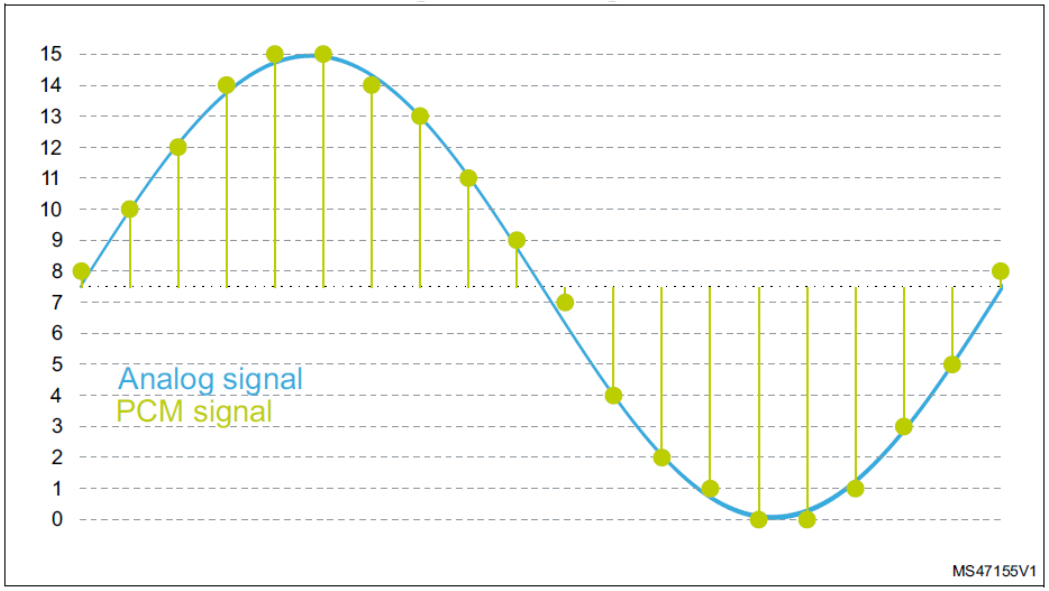
- You need headphones or a speaker compatible with 3.5mm audio jack.



PDM and PCM Signals

- Need to convert Pulse Density Modulation (PDM) signals acquired from microphone to Pulse Code Modulation (PCM) signals.





I^2S

- I^2S is a synchronous serial communication interface.
- It support four different audio standards including the I^2S Philips standard, the MSB- and LSB-justified standards, and the PCM standard.
- It can operate as a slave or a master device in full-duplex mode or in half-duplex mode (as a transmitter or a receiver). Master clock can be provided by the interface to an external slave component when the I^2S is configured as the communication master.
- Two instances of I^2S are used: 1) communicate with microphone (as a receiver) and 2) communicate with audio DAC (as a transmitter).
- It offers Direct Memory Access (DMA) capability for data transmission and reception.
- Details are in “STM32F407 Reference Manual: Serial Peripheral Interface (SPI)”.

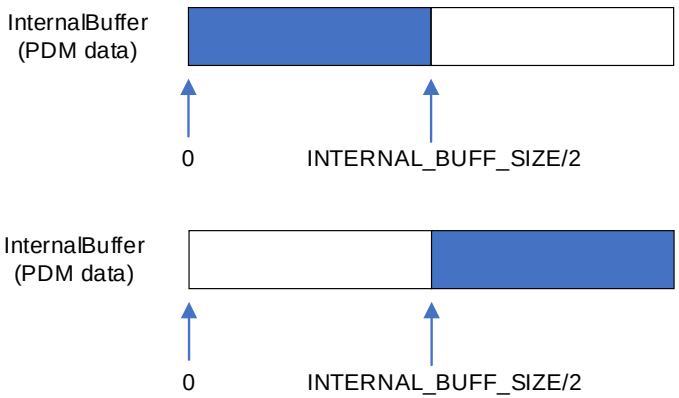
Direct Memory Access (DMA)

- Direct memory access (DMA) is used to provide high-speed data transfer between peripherals and memory, and between memory and memory.
- Data can be quickly moved by DMA without any CPU action. This keeps CPU resources free for other operations.

How to use DMA?

- **Initialize DMA.**
- **Start DMA.**
- **Implement interrupt callback functions to retrieve/reload data.**
  - DMA interrupts can be produced when
    - Half-transfer is reached.
    - Transfer is complete.
- **Details are in “STM32F407 Reference Manual: DMA Controller”.**

Microphone to Memory DMA



Initialize: BSP_AUDIO_IN_Init()

Start: BSP_AUDIO_IN_Record()

Callbacks:

- When the first half of buffer is filled with microphone PDM data,
  - BSP_AUDIO_IN_HalfTransfer_CallBack() is called.
- When the second half of buffer is filled with microphone PDM data,
  - BSP_AUDIO_IN_TransferComplete_CallBack() is called.
- In the callback functions, process the data.

Memory to Audio DAC DMA

- **Initialize: BSP_AUDIO_OUT_Init()**
- **Start: BSP_AUDIO_OUT_Play()**
- **Callbacks:**
  - No need to implement, let audio DAC continuously play the circular buffer.
- **Need to put processed data in the circular buffer to be played.**

Timing Considerations

- Microphone and I^2S are set up such that, every 1ms, one callback function (half or complete transfer) is called, PDM data is converted to PCM data, 16 samples of 16-bit of microphone audio data becomes available for processing. Therefore, the input sampling rate is 16K samples per second.
- Within 1ms, all signal processing needs to done to meet the real time constraint, including converting PDM to PCM and filter computations.
- The processed audio is saved into AudioOutBuffer to be played by audio DAC at 16K samples per second.

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